

Grosvenor Clocking Integration

User Guide

Access Limited Distribution

Workforce Management

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Document Control

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1. Overview

1.1. Purpose of the Guide

XD has teamed up with Grosvenor to integrate the clockings system. This guide aims to provide a high level overview of the system provided. For further detailed information see

<https://api.corehr.com>

1.2. Types of Integrations

The Grosvenor Integration consists of the following core areas:

- Employee Badge Management
- Badge Management
- Clock Times Management
- Transaction flow from Grosvenor to the XD platform
- Accessing employee data from XD and pushing to clocks
- Badge Web Service
- Configuration of terminal groups and endpoints

2. How it Works

2.1. Authentication

The authentication for clockings is based on OAuth 2 Client Credentials. To authenticate, all third-party applications require a Client ID and Client Secret which is obtained by contacting the tenant's administration.

Using the Client ID and Secret, a request is made to the authentication token end point and an access token is provided on successful authentication. This access token is required for each request and is included as part of the request header.

 **Note:** All requests to api.corehr should be SSL enabled

Further details on the authentication framework can be found under the x-Authentication section at <https://api.corehr.com>

2.2. Success and Error Codes

The table below details the codes which are returned, highlighting both successful functions as well as any errors which may be generated when the clockings are sent to XD.

Code ID	Explanation
Code 200	This code indicates that the file was successfully accepted.
Code 400	This code means there is an issue, and that Grosvenor will attempt to reload the file.
Code 500	This is considered a sustained functional error and the system will not deliver to that end for a period before trying to reload later.

3. Configuring Web Service Application Setup

3.1. Configuration

The following steps will need to be completed to configure the Web Service Application Set up.

1. Connect to XD Web Services.
2. Activate the Web Services the integration needs.

 **Note:** This step should be completed by the client to grant access to the relevant Web Services.

3.2. Webservice Application Setup Screen

1. Quick jump to the **Web Service Application Setup** screen
2. Click **Add Application** and enter the Application Name, Contact email and Application Description.
3. From the Application created, select **Update Privileges** from the 3-dot menu.
4. Enable the following Web Services:

Web Services
/v1/clocking/batch/clocks POST
/v1/clocking/groups GET
/v1/clocking/badge GET
/v1/image/user/:person GET

5. Navigate back to the Web Service Application Setup Widget.
6. From the Application created, select **View Credentials** from the 3-dot menu.
7. Click **Unmask Credentials**.
8. Provide Application credential details to integrator. For further details see <https://api.corehr.com>

4. Clocking Web Services

4.1. Key Points

Key points regarding the set up and function of the Clocking Web Services are outlined in the table below.

Function	Details
Formats & Interfaced Data	Formats and interfaced data should be agreed in advance between the client and XP Project Team.
Authentication Framework	Details on the authentication framework can be found under the x-Authentication section at https://api.corehr.com once they have been removed.
Web Service Integration	Web service integration is validated and signed off through user acceptance testing and parallel runs before being applied to production.
Badge Number	Badge numbers are assigned to personnel in advance, where there is integration by badge number.
Transferred Details	It is recommended that transferred details are also reviewed by viewing the details on the employee's timesheet or Manager Team Scheduler options. Errors can be viewed on the <i>Clocking Admin Screen</i> .
Attendance	The system recommends that Attendance is generated from the clocking data.
Temporary Data	Temporary data is cleared down prior to live running.
Multi Appointment	The clocking web service does not support multi appointment or clockings by appointment.

4.1.1. Badge Number

Within XD, all the clocking references are stored under the personnel number. The default option is always set as *Interface by badge number*.

The XD system reads the badge number from the interface. It retrieves the associated person number from the badge or person reference table and inserts the swipe date and time into the clocking table. This sets against the relevant personnel number.

**Note:**

- A person can only have one valid Active badge number at any given time.
- The badge number is a numeric only field, between 1 and 18 digits in length. The assignment of a badge number to an employee must exist in XD for the interface to successfully assign the correct personnel number to a clocking transaction.
- Badge numbers and Person numbers can be the same number.

4.1.2. Person Number Not Standard

When using Interfacing by Person Number, the requirement to store the badge or person number mapping table is removed.

To facilitate Clocking Integration by Person Number the following items should be completed in advance:

Item	Details
Format	There should be consultation between the client and the XD project team to agree the format of data being transferred.
Number of Characters	If transferring by person number, the number should not exceed 10 characters.

4.1.3. Batch Load Clocks into the XD System

For the best results, it is recommended to send less than 10,000 clockings per batch. If this amount is exceeded, the connection may time out. For a batch load of 10,000, the estimated time taken by the system is 3 minutes.

If there are any errors in data parse, they will be returned immediately to Grosvenor with a 400

error ([See Errors table](#)) and a response explaining why they were rejected.

Further details on format and the Clocking File are available on <https://api.corehr.com>

4.1.4. Admin Screen

The Clocking Admin Interface Screen is available to view the Clock batches that have loaded into the XD System and reviews any issues with the clocking loads.

To access the Clocking Admin Interface Screen:

1. Navigate to the **WFM Admin Dashboard**.
2. Click **WFM Configuration**.
3. Select **Admin Clocking Interface**.
4. The Clocking Interface screen displays.

Batch ID	External Batch ID	Stage	Processed Date	Total Records	Total Records Inserted	Total Record Errors
2021021801368167	123703	Processing Complete	18-Feb-2021 09:34	1	1	0
2021021131919185	123702	Processing Complete	11-Feb-2021 12:00	2	0	1
2021021131919183	123701	Processing Complete	11-Feb-2021 11:59	2	1	1
2021021131919181	123700	Processing Complete	11-Feb-2021 11:14	4	0	1
2021021131919179	123701	Processing Complete	11-Feb-2021 11:14	4	0	1

Process ID	External Batch ID	Stage	Processed Date	Total Records	Total Records Inserted	Total Record Errors
	123703	Ready to Process	18-Feb-2021 09:34	1		
2021021801368168		Processing Started	18-Feb-2021 09:34	0	0	0
2021021801368168		Processing Complete	18-Feb-2021 09:34	1	1	0

The Admin Clocking Interface screen displays the following fields:

Field	Explanation
Valid Loads	Details the loads which have been validated by the system.
Errors on Load	Details any errors which are found on the system.
Number of Records loaded	This gives you the number of clockings which have been loaded on the system.
Ability to rerun a batch	This allows you to rerun a batch of clockings.

Field	Explanation
View details of a batch reload	This allows you to look through any errors which need to be checked.
Days back to load data	This allows you go back 180 days
Email address	An email address should be provided to send errors to. If no email is provided no errors will be sent to an administrator
Frequency of error mails	This defaults to 0 which sends a mail every time an error occurs. Users can change this in multiples of hours e.g. 2 hours will only send a mail every two hours from the last successful run.
Frequency of failure mails	Users can decide the frequency that failure mails should be sent. This defaults to 0 which sends a mail every time a failure occurs. Users can change this in multiple of hours e.g. 2 hours will only send a mail every two hours from the last successful run.

To set up the Clocking Interface Admin Screen the following Menu Options and Web Services are required:

Menu Option	Menu ID	System
Admin Clocking Interface	CONFIG_SET_MENU.ADMIN_BATCH_CLOCKING	CorePortal
Get admin run	WS_CLOCKINGS_MENU.GET_ADMIN_RUN	Web Services
Create admin batch	WS_CLOCKINGS_MENU.POST_ADMIN_BATCH_ID	Web Services
Get admin batch	WS_CLOCKINGS_MENU.GET_ADMIN_BATCH	Web Services
Get admin batch error	WS_CLOCKINGS_MENU.GET_ADMIN_BATCH_ERROR	Web Services
Get admin settings	WS_CLOCKINGS_MENU.GET_ADMIN_SETTINGS	Web Services
Update admin settings	WS_CLOCKINGS_MENU.PATCH_ADMIN_SETTINGS	Web Services

5. Distribution Groups

Distribution Groups are groups of clockings that are sent in batches to the system. Once it has been configured, the system pulls employee data on a company-by-company basis from endpoints provided by XD.

5.1. First Synchronization

On first synchronization, the system pulls the list of Clock Groups or Distribution Groups. The list of Employees is pulled from PeopleXD which stores these users and calculates the list of employees for each clock based on distribution group membership. The clock is then sent Create, Update or Delete commands as necessary for each Employee.

For each user, a profile picture should be downloaded if available. PeopleXD uses etags to allow change detection on the server.

5.2. Distribution

For each configured company, the system, on the defined schedule pulls a complete list of Distribution Groups. This is done on a company-by-company basis from the endpoints provided by PeopleXD.

To access Clocking Distribution Groups:

1. Navigate to **WFM Admin Dashboard**
2. Click **WFM Configuration**
3. Select **Clocking Distribution Groups**
4. Click **Add**
5. Enter a **Group Code** and **Description**
6. PeopleXD always sends a full list of Distribution Groups as part of the Web Service and Grosvenor assigns out the distribution groups to the clocking devices.

5.3. Format

The distribution group is made up of code and description that is set out in the PeopleXD.

Format and further details on the badge detail file are available on <https://api.corehr.com>

5.4. Configuration

Clocking Distribution Groups can be assigned to an employee via the *Badge Details Screen*. The following menu options and web services are required:

Menu Option	Menu ID	System
Add/Delete Distribution Groups	PERSON_MAINT_ADM_MENU.CLOCKING_DISTRO_GROUPS	CorePortal
Add/Delete Distribution Groups	PERSON_MAINT_MGR_MENU.CLOCKING_DISTRO_GROUPS	CorePortal
Distribution Groups	CONFIG_SET_MENU.ADMIN_DISTRO_GROUPS	CorePortal
Create distribution	WS_CLOCKINGS_MENU.POST_DISTRIBUTION	Web Services
Create distribution person	WS_CLOCKINGS_MENU.POST_DISTRIBUTION_PERSON	Web Services
Create distribution person Proxy (Entire Organization)	WS_CLOCKINGS_MENU.POST_DISTRIBUTION_PERSON_PROXY	Web Services
Delete distribution	WS_CLOCKINGS_MENU.DELETE_DISTRIBUTION	Web Services
Delete distribution person	WS_CLOCKINGS_MENU.DELETE_DISTRIBUTION_PERSON	Web Services
Delete distribution person Proxy (Entire Organization)	WS_CLOCKINGS_MENU.DELETE_DISTRIBUTION_PERSON_PROXY	Web Services
Update distribution	WS_CLOCKINGS_MENU.PATCH_DISTRIBUTION	Web Services

6. Badge Web Service

6.1. Incremental Synchronisation

With Incremental Synchronization the list of Employees *changes* is pulled from XD. The list is compared to the data from the last successful sync operation and calculated with the updated list of users for each clock based on distribution group membership.

The clock is then sent Create, Update or Delete commands as necessary for each Employee. For newly imported users download profile picture if available.

6.2. Badge Details

The default synchronisation period is configurable, with a minimum time between synchronisation of 1 hour, there is no maximum. When the system updates employee images, the e tag of the image should be used for change detection.

A full synchronization should take place once a week to correct any drift that has occurred due to synchronization errors.